NATURAL LANGUAGE PROCESSING

ASSIGNMENT – 04

CO 4 – Semantic Analysis Through Structured Reasoning

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PART I – SEMANTIC ANALYSIS THROUGH STRUCTURED REASONING

Task 1 – Meaning Representation and Language Structure (8 Marks)

1.1 Identify Meaning Components (2 Marks)

Sentence: “Maya gave Ravi a book yesterday.”

Component	Identification
Entities	Maya (giver), Ravi (recipient), a book (thing transferred).
Event	GIVING; a transfer event.
Agent	Maya – the person who performs the giving.
Recipient	Ravi – the person who receives the book.
Theme	A book – the item transferred.
Time	Yesterday – the event happened on the day before the speaking day.
Linguistic form	A declarative English sentence with subject–verb–indirect object–direct object–time expression.
Represented meaning	A giving event with Maya as agent, Ravi as recipient, a book as theme, and yesterday as its time.

The surface form is the order and grammatical structure of the words. The meaning representation records the event and the semantic roles independently of the exact wording.

1.2 Compare Meaning Representations (3 Marks)

(a) Conceptual/event structure:

GIVE(e1) | Agent(e1, Maya) | Recipient(e1, Ravi) | Theme(e1, Book1) | Time(e1, Yesterday)

Here e1 identifies the giving event and Book1 is an individual object whose type is BOOK. This structure makes the event and its participants explicit.

(b) Frame / slot representation:

Frame: GIVING_EVENT	Value
Event ID	e1
Event type	Giving
Agent	Maya
Recipient	Ravi
Theme	a book
Time	yesterday

The event representation is compact and relation-focused. The frame is easy for an application to store because each role has a named slot. Both leave details such as the book title, exact date, location and manner unspecified.

1.3 Meaning Structure and Ambiguity (2 Marks)

Word order helps identify grammatical functions, but semantic roles identify the participant's contribution to the event. A passive alternative is: "A book was given to Ravi by Maya yesterday."

Its representation remains GIVE(e1), Agent(e1,Maya), Recipient(e1,Ravi), Theme(e1,Book1), Time(e1,yesterday). The surface subject changes from Maya to a book, but the semantic roles remain the same. Replacing "Ravi" with "Maya" as recipient would change the meaning.

1.4 Design Defence (1 Mark)

This representation is suitable for a computer because it stores meaning as named entities, events and role-value relations. A program can search, compare, validate or reason over fields such as Agent, Recipient and Time. It does not depend only on the original word order. Missing roles can be detected, and the same event can be represented even when the sentence is rewritten in passive voice. It is therefore structured data rather than a simple paraphrase.

Task 2 – First Order Predicate Calculus (8 Marks)

2.1 Define the Logical Vocabulary (2 Marks)

Symbol	Type / argument order	Meaning
Student(x)	Unary predicate	x is a student.
Course(c)	Unary predicate	c is a course.
Instructor(i)	Unary predicate	i is an instructor.
Enrolled(x,c)	Binary predicate	student x is enrolled in course c.
Teaches(i,c)	Binary predicate	instructor i teaches course c.
Passed(x,c)	Binary predicate	student x passed course c.
Rao	Constant	The individual Dr. Rao.
x, c, i	Variables	Range over individuals, courses and instructors as restricted by predicates.

Assumption: the domain contains people and courses; unary predicates distinguish their types. “At least one” means there exists one or more courses.

2.2 Translate Natural Language into Logic (3 Marks)

(i) Every student is enrolled in at least one course:

$\forall x\,[\text{Student}(x) \rightarrow \exists c\,(\text{Course}(c) \wedge \text{Enrolled}(x,c))]$

(ii) Some student passed every course taught by Dr. Rao:

$\exists x\,[\text{Student}(x) \wedge \forall c\,((\text{Course}(c) \wedge \text{Teaches}(\text{Rao},c)) \rightarrow \text{Passed}(x,c))]$

(iii) No instructor teaches every course:

$\neg \exists i\,[\text{Instructor}(i) \wedge \forall c\,(\text{Course}(c) \rightarrow \text{Teaches}(i,c))]$

Equivalent form for (iii): $\forall i\,[\text{Instructor}(i) \rightarrow \exists c\,(\text{Course}(c) \wedge \neg\text{Teaches}(i,c))]$.

2.3 Interpret and Check the Formulae (2 Marks)

For (i), the outer universal quantifier $\forall x$ applies to every individual x. If x is a student, the implication requires at least one course c for which Course(c) and Enrolled(x,c) are true. The existential quantifier is inside the scope of $\forall x$, so different students may be enrolled in different courses.

Quantifier order can change meaning. Compare $\forall x \exists c$ Enrolled(x,c) ("each student has some course") with $\exists c \forall x$ Enrolled(x,c) ("there is one course in which every student is enrolled").

Finite model	Facts	Result
Students: Ana, Ben; courses: NLP, AI	Enrolled(Ana,NLP), Enrolled(Ben,AI)	$\forall x \exists c$ is true; $\exists c \forall x$ is false.

Therefore the formulae are not logically equivalent. The example is a countermodel to equivalence.

2.4 Logical Consistency (1 Mark)

The pair cannot both be true if "NLP" refers to the same course and the domain is classical logic. "Every student passed NLP" means $\forall x(\text{Student}(x) \rightarrow \text{Passed}(x,\text{NLP}))$. "There is a student who did not pass NLP" means $\exists x(\text{Student}(x) \wedge \neg\text{Passed}(x,\text{NLP}))$. The second statement supplies a student who makes the first statement false.

Task 3 – Linguistically Relevant Concepts and Syntax-Driven Semantic Analysis (9 Marks)

3.1 Identify Linguistically Relevant Concepts (2 Marks)

Sentence: "The experienced researcher saw the student with the telescope."

Concept	Example / role	Why it matters
Entities	researcher; student; telescope	Identifies participants and objects.
Event / predicate	SEE(e)	Represents the seeing action.
Agent / experiencer	experienced researcher	Person who sees.
Theme / perceived entity	student	Person being seen.
Modifier	experienced	Describes the researcher.
Prepositional phrase	with the telescope	Can attach to the seeing event or to the student noun phrase.

Semantic relations	Agent(e,researcher), Theme(e,student), Instrument(e,telescope) or With(student,telescope)	Distinguishes the two readings.

3.2 Syntax-to-Semantics Mapping (3 Marks)

Parse A – instrument attachment to the verb phrase:

[S [NP The experienced researcher] [VP saw [NP the student] [PP with the telescope]]]

Meaning: the researcher used a telescope to see the student.

SEE(e1) ∧ Agent(e1,researcher) ∧ Theme(e1,student) ∧ Instrument(e1,telescope) ∧ Experienced(researcher)

Parse B – modifier attachment to the noun phrase:

[S [NP The experienced researcher] [VP saw [NP the student [PP with the telescope]]]]

Meaning: the researcher saw a student who had or was carrying a telescope.

SEE(e1) ∧ Agent(e1,researcher) ∧ Theme(e1,student) ∧ With(student,telescope) ∧ Experienced(researcher)

The parse tree determines whether PP “with the telescope” combines with VP saw or with NP student. That attachment choice changes the semantic relation.

3.3 Semantic Composition (2 Marks)

1. “researcher” creates an entity of type PERSON.
2. “experienced” adds the property Experienced(researcher).
3. “the student” creates the entity student and its person/student properties.
4. “saw” creates a SEE event with the researcher as Agent and student as Theme.
5. “with the telescope” adds either Instrument(SEE-event,telescope) or With(student,telescope), depending on the parse.

The interpretations diverge at the PP attachment step, not at the meanings of “researcher,” “student,” or “saw.”

3.4 Failure Analysis (2 Marks)

- Error 1: The parser attaches the PP to the wrong phrase, so the system says the student owns the telescope when it was actually used by the researcher. Validation: check the parse alternatives and compare instrument-role compatibility with the verb SEE.
- Error 2: The system assigns Agent and Theme using word order alone. In passive or unusual constructions this can reverse the roles. Validation: use grammatical dependencies and verb-role constraints; confirm that the subject/object functions match the selected predicate frame.

Task 4 – Lexemes, Senses and Word Sense Disambiguation (9 Marks)

4.1 Lexemes and Senses (2 Marks)

A lexeme is an abstract vocabulary item with one or more related senses. A word form is the actual written or spoken form in a sentence. A sense is one particular meaning of the lexeme.

Context	Word form / POS	Lexeme and sense
Nila deposited cash at the bank.	bank – noun	$BANK_1$: financial institution.
They sat on the bank of the river.	bank – noun	$BANK_2$: land beside a river.
The pilot banked the aircraft to the left.	banked – verb, past tense	$BANK_3$: tilt an aircraft sideways during a turn.

4.2 Design a WSD Method (3 Marks)

Input: a sentence containing bank/banked. Output: a sense label BANK_FINANCIAL, BANK_RIVER_EDGE or BANK_AIRCRAFT_TILT.

Candidate senses are the three senses above. Use a context window of three words on each side, the part of speech, and trigger words.

Evidence / feature	Sense supported
cash, deposit, money, account, withdraw, ATM	BANK_FINANCIAL
river, water, shore, sit, edge, muddy	BANK_RIVER_EDGE
pilot, aircraft, plane, wing, turn, left	BANK_AIRCRAFT_TILT

Decision rule: first identify part of speech. If the word is a verb and aircraft/pilot/wing/turn occurs, choose BANK_AIRCRAFT_TILT. If it is a noun, compare financial and river trigger counts in the

context window. Select the sense with the highest count. If scores tie or no trigger appears, return UNKNOWN/AMBIGUOUS rather than guessing.

4.3 Test the Method (2 Marks)

Sentence	Evidence	Output
Nila deposited cash at the bank.	deposited, cash	BANK_FINANCIAL
They sat on the bank of the river.	river, sat	BANK_RIVER_EDGE
The pilot banked the aircraft to the left.	pilot, aircraft, left; verb form	BANK_AIRCRAFT_TILT
We opened a savings account at the bank.	savings, account	BANK_FINANCIAL

Possible failure: “The pilot walked along the river bank.” Both pilot and river occur, but “river” and “along” support the river-edge sense. A simple keyword count may be confused if it gives too much weight to pilot. A better method uses dependency relations and phrase-level context.

4.4 Evaluation (2 Marks)

Create a labelled test set of 30 sentences: 10 financial, 10 river-edge and 10 aircraft-tilt examples. Two annotators assign the correct sense; disagreements are reviewed and labelled as ambiguous if context is insufficient. Compare system predictions with gold labels.

Accuracy = correct predictions / total test instances × 100. Example: 27 correct out of 30 gives 27/30 × 100 = 90%. A balanced test set gives equal weight to each sense, but real text may contain mostly financial uses. Accuracy on a balanced set may therefore not reflect real-world performance. Report per-sense precision/recall or a confusion matrix as well.

Task 5 – Information Retrieval and Semantic Matching (8 Marks)

5.1 Query and Document Representation (2 Marks)

Query: “methods for finding the meaning of ambiguous words”. Central concepts: word meaning, ambiguity, context, sense identification and WSD methods. Intent: retrieve explanations or methods that identify which sense of a word is intended in a particular context.

Represent the query as concept terms: {word, meaning, ambiguity, context, sense, disambiguation, WSD}. Represent each document with title terms, body terms, topic concepts and an embedding/vector for semantic similarity.

5.2 Lexical versus Semantic Retrieval (2 Marks)

Keyword retrieval matches exact or similar words. It may miss a document titled “Choosing the correct interpretation of polysemous expressions” because it does not contain the exact phrase “ambiguous words.” Semantic retrieval can connect polysemous expressions with ambiguous words and interpretation with meaning selection.

5.3 Ranking Design (2 Marks)

Use a score: Score(Q,D) = 0.30L + 0.50S + 0.20C, where L is normalized lexical overlap, S is semantic similarity, and C is concept coverage. Each signal is between 0 and 1. Semantic similarity receives the highest weight because the query asks for meaning-based methods. Sort by score from highest to lowest; break ties using concept coverage, then document quality.

Document	Description	L	S	C	Score
D1	Introduction to word sense disambiguation	0.90	0.95	0.90	0.925
D2	Contextual interpretation of polysemous words	0.35	0.92	0.90	0.775
D3	Dictionary list of common words	0.55	0.30	0.35	0.385
D4	How to detect ambiguous words using context	0.75	0.88	0.85	0.840
D5	History of library book classification	0.25	0.20	0.20	0.215

Ranking: D1 (0.925), D4 (0.840), D2 (0.775), D3 (0.385), D5 (0.215). D2 can rank above a literal keyword document because its concepts match the intended task even though its wording differs.

5.4 Evaluation and Error Analysis (2 Marks)

Assume D1, D2 and D4 are relevant. At top 3, the system retrieves D1, D4 and D2: relevant retrieved = 3; retrieved = 3; total relevant in collection = 3.

Precision = relevant retrieved / all retrieved = 3/3 = 1.00 (100%). Recall = relevant retrieved / all relevant documents = 3/3 = 1.00 (100%).

A semantic error could return a general dictionary document because it discusses word meanings but does not explain disambiguation. This is a false positive. Improve ranking by adding a task-intent feature that rewards documents describing context-based sense selection and evaluation.

Task 6 – Integrated Semantic Analysis Challenge (8 Marks)

Stage 1 – Interpret the Problem (1 Mark)

Sentence: “I saw her duck near the bank, and it was closed.” Ambiguities: “duck” can be a noun (bird) or verb (lower her head/body); “bank” can be a financial institution or river edge; “it” may refer to bank or another singular object. “Closed” strongly fits a bank as a business, but context is still needed.

Stage 2 – Build Candidate Interpretations (2 Marks)

Candidate A: I saw her duck (the bird) near the bank (financial institution), and the bank was closed.

Representation: SEE(e1) ∧ Agent(e1,I) ∧ Theme(e1,DuckBird) ∧ Near(DuckBird,BankInstitution) ∧ Closed(BankInstitution).

Candidate B: I saw her duck (lower her head) near the bank (river edge), and the river bank/nearby facility was closed. This reading is less natural because “closed” does not normally describe a river bank.

Representation: SEE(e1) ∧ Agent(e1,I) ∧ Theme(e1,Her) ∧ Action(e1,DUCK) ∧ Near(e1,RiverBank) ∧ Closed(RiverBank).

The second reading requires a special context, such as a fenced or closed riverside area. Another possible reading is that “her duck” is the bird and “bank” is the river bank; the final pronoun it then refers to a separate closed location only if the discourse supplies one.

Stage 3 – Syntax-Driven Analysis and WSD (2 Marks)

The parser identifies “her duck” as a noun phrase if duck is a noun, or analyses “duck” as a verb with “her” as its object if the construction permits that reading. The noun phrase “near the bank” modifies the seeing event or the duck. The pronoun “it” is resolved by agreement and discourse salience. “Closed” is a strong selectional clue for a business or a facility, not an ordinary river edge.

Useful additional context: Was the speaker trying to withdraw money, walking beside a river, watching a bird, or describing a restricted area? If evidence remains weak, retain both readings and ask: “Do you mean a financial bank or the side of a river?”

Stage 4 – Information Retrieval Trace (2 Marks)

If context indicates a financial institution, query concepts are: financial bank + opening hours/status + closed branch. Candidate articles:

Article	Content	Sense match	Intent match	Decision
A1	Why a bank branch may be closed	Financial	High	Rank 1
A2	How to identify ducks near wetlands	Bird	Low	Rank 2 only if duck is intended
A3	Riverbank access and closure notices	River edge	Medium only with river context	Rank lower
A4	How to close a bank account	Financial	Different meaning of closed	Distractor

The retrieval system first filters by the selected BANK sense and then ranks by query intent. A4 contains “bank” and “close” but describes closing an account, not checking whether a branch is closed. Sense compatibility and intent matching prevent it from being selected as the best result.

Stage 5 – Challenge and Refine (1 Mark)

Misleading input: “I saw her duck near the bank after the flood; it was closed.” The words “flood” and “river” may support the river-bank sense, while “closed” may refer to a restricted riverside path. A

financial-only model may fail. Refine it by using nearby discourse, event context, pronoun links and a confidence threshold. If confidence is low, show alternatives or ask a clarification question.

PART II – SEMANTIC SYSTEMS ENGINEERING

Task 1 – Representing Meaning and Language Structure (8 Marks)

1.1 Semantic Structure Analysis (2 Marks)

Sentence: "Leena sent the report to Arun after the meeting."

Element	Meaning
Event	SEND(e1) – a transfer/communication event.
Agent	Leena – person who sends.
Theme	The report – item sent.
Recipient	Arun – intended receiver.
Temporal expression	After the meeting – sending occurs later than the meeting.
Implicit information	The exact time, delivery method, report contents and meeting identity are not specified.

Surface form: subject + verb + object + recipient phrase + time phrase. Represented meaning: a SEND event with Agent, Theme, Recipient and a temporal relation AFTER(e1, meeting1).

1.2 Build Meaning Representations (3 Marks)

(a) Semantic-role/event frame:

SEND(e1); Agent(e1,Leena); Theme(e1,Report1); Recipient(e1,Arun); After(e1,Meeting1).

(b) Graph / slot representation:

Node / relation	Value
Event node	e1 : SEND
Agent edge	e1 → Leena
Theme edge	e1 → Report1
Recipient edge	e1 → Arun
Temporal edge	e1 → AFTER(Meeting1)

Entity types	Leena: PERSON; Report1: DOCUMENT; Arun: PERSON; Meeting1: EVENT

The event frame gives named semantic roles. The graph shows how the event is connected to entities and the meeting. Both make the participant relationships explicit.

1.3 Meaning-Preserving Variation (2 Marks)

Passive rewrite: "The report was sent to Arun by Leena after the meeting." The grammatical subject changes, but Agent=Leena, Theme=report, Recipient=Arun and the AFTER relation remain unchanged. A meaning-changing rewrite would be "Leena sent Arun's report to the meeting," because it changes the recipient/theme relations.

1.4 Representation Check (1 Mark)

Validation rule: every SEND event must have exactly one identified Agent and at least one Theme. If Agent or Theme is missing, flag the representation as incomplete. Also check that the Recipient is a person or an appropriate receiving organization.

Task 2 – First Order Predicate Calculus: Formalise and Test (9 Marks)

2.1 Predicate Vocabulary (2 Marks)

Symbol	Argument order	Meaning
Learner(x)	x	x is a learner.
Course(c)	c	c is a course.
Enrolled(x,c)	learner, course	x is enrolled in c.
Completed(x,c)	learner, course	x completed c.
Prerequisite(p,c)	prerequisite course, target course	p is a prerequisite for c.
NLP	constant	The NLP course.
DeptCourse(c)	c	c is offered by the department.

Variables x,y range over learners; c,p range over courses. DeptCourse is introduced to represent "offered by the department."

2.2 Formalisation (3 Marks)

(i) Every learner enrolled in NLP has completed at least one prerequisite:

$\forall x\,[(\text{Learner}(x) \wedge \text{Enrolled}(x,\text{NLP})) \rightarrow \exists p\,(\text{Course}(p) \wedge \text{Prerequisite}(p,\text{NLP}) \wedge \text{Completed}(x,p))]$

(ii) Some learner completed every course offered by the department:

$\exists x\,[\text{Learner}(x) \wedge \forall c\,((\text{Course}(c) \wedge \text{DeptCourse}(c)) \rightarrow \text{Completed}(x,c))]$

(iii) No learner completed a course that they were not enrolled in:

$\forall x \forall c\,[(\text{Learner}(x) \wedge \text{Course}(c) \wedge \text{Completed}(x,c)) \rightarrow \text{Enrolled}(x,c)]$

2.3 Quantifier-Scope Challenge (2 Marks)

Compare $\forall x \exists c$ Enrolled(x,c) with $\exists c \forall x$ Enrolled(x,c). The first says every learner has at least one course, possibly a different course for each learner. The second says one particular course is shared by all learners.

Model	Facts	Truth values
Learners A and B; courses NLP and AI	Enrolled(A,NLP), Enrolled(B,AI)	$\forall x \exists c$ is true; $\exists c \forall x$ is false.

The two statements are not logically equivalent because this finite model makes one true and the other false.

2.4 Consistency Test (2 Marks)

Knowledge base facts: Learner(Ana); Learner(Ben); Course(NLP); Course(AI); Enrolled(Ana,NLP); Completed(Ana,NLP); Enrolled(Ben,AI). Rule: $\forall x \forall c\,[(\text{Learner}(x) \wedge \text{Completed}(x,c)) \rightarrow \text{Enrolled}(x,c)]$.

Query: Enrolled(Ana,NLP)? Yes. It follows directly from the stored fact. Query: Completed(Ben,AI)? No; it is not present and cannot be derived from the rule. Under an open-world assumption, absence of a fact does not prove its negation. Under a closed-world database assumption, it may be treated as unknown/not recorded, not necessarily false.

Task 3 – Linguistic Concepts and Syntax-Driven Semantic Analysis (9 Marks)

3.1 Linguistically Relevant Concepts (2 Marks)

Sentence: “The analyst noticed the student with the camera near the library.”

Concept	Possible value
Entities	analyst, student, camera, library
Predicate/event	NOTICE(e1)
Agent/experiencer	analyst
Theme	student
Modifiers	the; possible descriptive phrase near the library
PP relations	with(camera) may modify student or notice event; near(library) may modify student or event
Attachment points	with-PP → NP student or VP noticed; near-PP → NP student or VP/event

3.2 Parse-to-Meaning Procedure (3 Marks)

Parse A – both phrases describe the student:

[VP noticed [NP the student [PP with the camera] [PP near the library]]]

Meaning: the analyst noticed a student who had a camera and was near the library.

NOTICE(e1) ∧ Agent(e1,analyst) ∧ Theme(e1,student) ∧ Has(student,camera) ∧ Near(student,library).

Parse B – phrases describe the noticing event:

[VP noticed [NP the student] [PP with the camera] [PP near the library]]

Meaning: the analyst used a camera while noticing the student near the library (or the noticing event occurred near the library).

NOTICE(e1) ∧ Agent(e1,analyst) ∧ Theme(e1,student) ∧ Instrument(e1,camera) ∧ Location(e1,library).

The first divergence occurs when “with the camera” attaches to the NP student versus the VP noticed. A second divergence occurs when “near the library” attaches to the student versus the event.

3.3 Composition and Type/Role Checks (2 Marks)

6. Build entity representations for analyst, student, camera and library.
7. Build NOTICE event with Agent=analyst and Theme=student.

8. Attach “with the camera” and assign either Has(student,camera) or Instrument(event,camera).
9. Attach “near the library” and assign either Near(student,library) or Location(event,library).

Check 1: semantic type check – an Instrument role should accept a usable instrument, such as a camera, while an Agent role should normally accept an animate entity. Check 2: attachment check – ensure each PP has a recorded syntactic head and that its semantic relation is compatible with that head.

3.4 Error Challenge (2 Marks)

A syntactically possible but implausible reading is that the analyst used the student as an instrument to notice something. A parser might produce this if it relies only on attachment patterns. The system should use selectional preferences: Agent and Theme of NOTICE are typically people or perceivable entities, while Instrument is typically a tool. Contextual evidence and type constraints should lower the implausible reading’s score.

Task 4 – Lexemes, Senses and Word Sense Disambiguation (8 Marks)

4.1 Lexeme and Sense Inventory (2 Marks)

Example	Word form / POS	Sense label and meaning
Turn on the light.	light – noun	LIGHT_NOUN_DEVICE: lamp or source of illumination.
This bag is light.	light – adjective	LIGHT_ADJ_LOW_WEIGHT: not heavy.
The light meal was served early.	light – adjective	LIGHT_ADJ_SMALL_OR_EASY: not substantial or heavy.

Lexeme: LIGHT. Word form: the actual form “light” in each sentence. Sense: the intended meaning in that context. The noun and adjective uses have different grammatical categories.

4.2 WSD Design (2 Marks)

Input: sentence containing “light”. Candidate senses: LIGHT_NOUN_DEVICE, LIGHT_ADJ_LOW_WEIGHT, LIGHT_ADJ_SMALL_OR_EASY. Features: part of speech, nearby words, dependency relation and collocations.

Rule / context clue	Predicted sense
POS=noun; verbs such as turn on, switch, turn	LIGHT_NOUN_DEVICE

off; words lamp, bulb	
POS=adjective; modifies bag, box, suitcase; words carry, weight, heavy	LIGHT_ADJ_LOW_WEIGHT
POS=adjective; modifies meal, snack, food; words eat, calories, dinner	LIGHT_ADJ_SMALL_OR_EASY

Decision procedure: identify POS; collect nearby nouns and verbs within a five-word window; apply the matching rule with the strongest evidence. If no rule matches or two rules tie, output AMBIGUOUS and retain candidate senses.

4.3 Apply and Challenge (2 Marks)

Sentence	Evidence	Output
Turn on the light.	turn on; noun phrase	LIGHT_NOUN_DEVICE
This bag is light.	bag; adjective after linking verb	LIGHT_ADJ_LOW_WEIGHT
The light meal was served early.	meal; served	LIGHT_ADJ_SMALL_OR_EASY
She packed a light suitcase for travel.	suitcase; packed	LIGHT_ADJ_LOW_WEIGHT

Failure case: “The light food container was easy to carry.” Here light may mean low in weight, while food may mislead a simple food-collocation rule toward the small/easy meal sense. Dependency structure shows that light modifies container, not food.

4.4 Evaluation (2 Marks)

Eight labelled test instances:

ID	Sentence	Gold label	Prediction
1	Turn on the light.	DEVICE	DEVICE
2	The lamp gives bright light.	DEVICE	DEVICE
3	This bag is light.	LOW_WEIGHT	LOW_WEIGHT
4	A light suitcase is easy	LOW_WEIGHT	LOW_WEIGHT

	to carry.		
5	We ate a light meal.	SMALL_MEAL	SMALL_MEAL
6	A light snack was enough.	SMALL_MEAL	SMALL_MEAL
7	The light box was easy to lift.	LOW_WEIGHT	LOW_WEIGHT
8	The light dinner included soup.	SMALL_MEAL	SMALL_MEAL

Annotation policy: label the sense expressed by the word in its local sentence context; if two meanings are genuinely possible, mark AMBIGUOUS and exclude it from single-label accuracy or report it separately. In this example all 8 predictions are correct.

Accuracy = 8/8 × 100 = 100%. This is only a small demonstration, not proof of real-world performance. If most examples are device uses, a system can obtain high accuracy by always predicting DEVICE. Report per-class scores and test on varied contexts.

Task 5 – Semantic Information Retrieval (8 Marks)

5.1 Query and Document Semantics (2 Marks)

Query: “How can a system identify the intended meaning of a word in context?” Intent: find methods for word sense disambiguation, contextual meaning selection and sense classification.

ID	Candidate document	Relevance
D1	A practical guide to word sense disambiguation	Relevant
D2	Using context to choose the correct meaning of polysemous words	Relevant
D3	A dictionary of common English words	Partly relevant / broad
D4	How to classify words by their grammatical category	Distractor
D5	Contextual language models for	Relevant

	selecting word meanings	
D6	How to identify words in scanned documents using OCR	Distractor

Represent query as concepts {word, intended meaning, context, sense, disambiguation, WSD}. Represent documents using title/body terms, concept labels and semantic vectors.

5.2 Ranking Strategy (2 Marks)

Use Score(Q,D) = 0.40L + 0.60S, where L is normalized lexical overlap and S is semantic similarity. Both values range from 0 to 1. Semantic similarity has greater weight because the query is about identifying meaning, not merely finding matching words. Break ties by higher concept coverage, then by a more specific title.

5.3 Retrieval Trace (2 Marks)

Document	L	S	Calculation	Final score
D1	0.90	0.95	0.40(0.90)+0.60(0.95)	0.930
D2	0.40	0.94	0.40(0.40)+0.60(0.94)	0.724
D3	0.60	0.35	0.40(0.60)+0.60(0.35)	0.450
D4	0.50	0.25	0.40(0.50)+0.60(0.25)	0.350
D5	0.35	0.90	0.40(0.35)+0.60(0.90)	0.680
D6	0.45	0.20	0.40(0.45)+0.60(0.20)	0.300

Ranking: D1, D2, D5, D3, D4, D6. D5 ranks above a literal keyword distractor because it discusses contextual models that select word meanings, which is close to the query intent.

5.4 Evaluate (2 Marks)

Relevant documents are D1, D2 and D5. At cutoff 3, all three are retrieved.

Precision@3 = 3 relevant retrieved / 3 retrieved = 1.00 (100%). Recall@3 = 3 relevant retrieved / 3 relevant in the collection = 1.00 (100%).

False positive example: D3 may be returned because dictionaries define words, although it may not explain contextual disambiguation. False negative example: a relevant article using the phrase “lexical ambiguity resolution” may be missed. Improve the system with concept expansion for terms such as

polysemy, sense selection and lexical ambiguity, and add a relevance check for whether the document explains a method.

Task 6 – Integrated Semantic Analysis Design Challenge (8 Marks)

6.1 Model the Ambiguity (1 Mark)

Sentence: “I watched the crane by the bank, then it moved.” Crane may mean a bird or a lifting machine. Bank may mean a financial institution or a river edge. “It” may refer to the crane or the bank. Useful context includes construction work, wetlands, water, money, movement type and the previous discourse.

6.2 Candidate Meaning Structures (2 Marks)

Candidate A – bird near a river bank:

Lexeme senses: CRANE_BIRD; BANK_RIVER_EDGE. Roles: watcher=I; watched-object=bird; location=river bank; moved-object=bird.

WATCH(e1) ∧ Agent(e1,I) ∧ Theme(e1,Bird1) ∧ Near(Bird1,RiverBank1) ∧ MOVE(e2) ∧ Theme(e2,Bird1).

Candidate B – lifting machine near a financial bank/building:

Lexeme senses: CRANE_MACHINE; BANK_FINANCIAL. Roles: watcher=I; watched-object=crane machine; location=near bank building; moved-object=crane machine.

WATCH(e1) ∧ Agent(e1,I) ∧ Theme(e1,CraneMachine1) ∧ Near(CraneMachine1,BankBuilding1) ∧ MOVE(e2) ∧ Theme(e2,CraneMachine1).

A third plausible reading is a crane machine near a river bank at a construction site. The sentence alone does not determine the senses.

6.3 Syntax and WSD Trace (2 Marks)

The parser identifies “the crane” as the object of watched and “by the bank” as a location phrase. The pronoun “it” is linked to a singular antecedent, most naturally the crane because it is the subject of the next event and can move. Context words such as wings, feathers, marsh or nest support the bird sense. Words such as construction, lifting, load, hook or operator support the machine sense. If the evidence is insufficient, keep both interpretations and ask whether the user means a bird or a lifting machine.

6.4 Semantic Retrieval (2 Marks)

Assume context indicates a construction site and crane machine. Query concepts: mobile construction crane + operation/movement + safety or technical information.

Article	Content	Sense compatibility	Rank
A1	How mobile construction cranes move and operate	Machine; high intent match	1
A2	Crane birds in wetland habitats	Bird; wrong sense	4
A3	Construction crane movement and site safety	Machine; high intent match	2
A4	River bank erosion and flood management	Bank river-edge; unrelated to crane machine operation	3

A2 shares the exact word “crane” but has the bird sense, so it is penalized by sense compatibility. A4 shares “bank” but does not answer the machine-operation query. Ranking uses sense match first, then concept similarity and query intent.

6.5 Adversarial Refinement (1 Mark)

Misleading context: “I watched the crane by the bank where birds gathered, then it moved.” “Birds” may cause the system to select the bird sense even if the crane is a machine at a construction site. The design may fail if it relies on one keyword. Refine it with multiple contextual features, dependency-based relations, confidence scoring and a clarification question when candidate scores are close.

Overall Conclusion

This assignment demonstrates how an NLP system moves from natural-language input to structured meaning. Semantic roles and event frames represent who did what, to whom, where and when. First-order predicate calculus expresses general statements and supports logical reasoning. Syntax-driven semantic composition handles phrase attachment, while word sense disambiguation selects meanings using grammatical and contextual evidence. Semantic retrieval ranks documents by both wording and

meaning. Testing with counterexamples, labelled examples, precision, recall and accuracy helps identify weaknesses. When context is insufficient, a reliable system should preserve alternatives or ask for clarification instead of making an unsupported guess.